

SAMA MOHAMED RASHAD

Electronics & Communications Engineering Student

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PROFESSIONAL SUMMARY

Electronics & Communications Engineering student at Tanta University with practical experience in embedded systems, microcontroller programming, and engineering simulation. Skilled in Python and Embedded C, with experience using Arduino, ESP32, and AVR microcontrollers. Currently expanding technical knowledge in computer networking and network infrastructure through ongoing training at NTI.

EDUCATION

Tanta University

2023 - present

Bachelor of Engineering in Electronics & Communications Engineering

Third-Year Student

TECHNICAL SKILLS

- **Programming:** Python, Embedded C
- **Embedded Systems:** AVR ATmega32, Arduino, ESP32, Microcontroller Interfacing
- **Engineering & Simulation:** MATLAB, Proteus, NI Multisim

PROJECTS

Smart Load Identification System

May 2026

Team Project | ESP32, Python, Bluetooth, Feature Extraction, Machine Learning

- Implemented real-time voltage and current data acquisition using ESP32 and electrical sensors.
- Established Bluetooth communication between the ESP32 and computer for real-time data transmission.
- Used Python to process and analyze collected electrical measurements and extract relevant electrical features.
- Applied an existing machine learning classification implementation to the collected dataset to evaluate load identification.

Sound Frequency Monitoring System

Dec 2025

Team Project | Arduino, MATLAB, Signal Processing, FFT Analysis

- Developed a real-time system for monitoring sound frequency and detecting abnormal variations through acoustic signal analysis.
- Integrated a microphone sensor with Arduino for real-time sound signal acquisition.
- Used MATLAB for signal visualization and frequency analysis, including FFT-based analysis and Zero-Crossing frequency estimation.
- Implemented visual and audible alert mechanisms using LED indicators and a buzzer for detected abnormal conditions.

Smart Lock System

Jul 2025

Individual Project | AVR ATmega32, Embedded C, Keypad, LCD, EEPROM, Proteus

- Designed and implemented a password-based smart lock system using the AVR ATmega32 microcontroller.
- Programmed the system in Embedded C, implementing keypad, LCD, EEPROM, motor, and buzzer interfacing.
- Developed security mechanisms including lockout periods and alarm activation after repeated incorrect password attempts.
- Simulated and tested the complete system using Proteus, including the door-opening mechanism.

Study Mate

May 2025

Team Project | Python, Tkinter, File Handling

- Developed the To-Do List module for a student productivity application using Python and Tkinter.
- Implemented task management functionality, including saving and loading tasks using file-based storage.
- Designed and developed the graphical user interface components for the task management section.

TRAINING

Network Infrastructure Training — NTI

Aug-Sep 2026

Summer Training | 120 Hours | Score: 98.5%

- Simulated and configured enterprise network topologies using core Routing & Switching concepts.
- Implemented network segmentation (VLANs) and efficient IP addressing schemes (Subnetting, DHCP).
- Configured static and dynamic routing protocols (e.g., OSPF) for optimal data flow.
- Deployed secure WAN connectivity (VPNs) and internet access services (NAT, PAT).

Microcontroller-Based Systems Training — NTI

Jun–Jul 2025

Summer Training | 120 Hours | Score: 89%

- Developed practical experience with AVR ATmega32 microcontrollers using Embedded C.
- Worked with microcontroller peripherals and interfacing, including timers, ADC, LCD, and keypad.

LANGUAGES

Arabic: Native | English: Intermediate